“Beneath the Surface: Safe Carbon Storage in India's Coal Basins”

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ABSTRACT:

India is one of the country with developing mixed economy requires huge amount of energy to complete daily energy needs. Alone, India's power sector is responsible for release of harmful greenhouse gases that has caused disturbances in the Earth's atmosphere. Combating climate change by reducing the levels of one of the most harmful greenhouse gases i.e. carbon dioxide through capture and storing in the coal seams would help in mitigating the climate change to some extent. Carbon dioxide is absorbed onto microporous structure of coal seams displacing methane molecules from their absorption sites. This process increases the production of low- carbon, eco- friendly fuel.

The main aim of carbon sequestration is focused on the underground coal of the Lower Gondwana sequence in Jharkhand state, India. Some of the best- fit model estimates the volumes of gases involved in CO2- enhanced coal bed methane. The best- fit model also provided analysis of gas distribution. Results indicates that CO2- enhanced CBM recovery is technically feasible in India.

Keywords: Coal seams, Coal bed methane, CBM recovery

INTRODUCTION:

Carbon capture and storage is the process in which carbon dioxide is extracted out at high temperatures using fossil fuel as primary fuel and then transported for storage. Carbon capture and storage is an efficient process to tackle the climate change making it more reliable. It's a good step to incorporate for a zero- emissions future. Coal plays a key role in fulfilling India's energy requirement. The economy is expected to grow at a rate of 7.4% in 2018 and thereby required more amount of energy to fulfill this need. The Energy Balancing Act states India is currently in a bit of a squeeze. Between 2005 and 2013, the country's hunger for power grew faster than its ability to provide it, leading to a "power deficit" (basically, the grid couldn't keep up with the demand).

The Current Scoreboard: The Heavyweights: Thermal power (mostly coal) does the heavy lifting, providing about 67% of our capacity. The Rising Stars: Renewables make up a smaller slice, with Wind Energy leading the pack at over half of all renewable electricity, The "Hidden" Fuel i.e. Coal isn't just a rock; it's like a sponge. Inside its tiny pores, it holds Coalbed Methane (CBM). Usually, this gas is a nightmare for miners—it's explosive and a massive contributor to global warming (20 times more potent than CO_2). But if we catch it before we mine the coal, it becomes a clean-burning fuel for homes and power plants. Why is the Government excited? They've rolled out the red carpet for investors to find this gas by: Cutting out the red tape (no upfront "signature" bonuses). This is where the science gets really cool. Imagine a coal seam underground. It naturally holds onto methane (CH_4). Through a process called Enhanced Coalbed Methane (ECBM), we can pump CO_2 (the stuff warming the planet) into the coal. Because coal "likes" CO_2 more than methane, a swap happens: The coal grabs the CO_2 and locks it away (Adsorption). The methane is pushed out so we can use it for energy (Desorption). For every 1 molecule of methane we get out, the coal can trap roughly 2 to 3 molecules of CO_2. It's like a storage locker for pollution that pays you back in fuel.

LITERATURE REVIEW

While giants like the USA and Australia have mastered this, India is still in the "learning to drive" phase. We have plenty of resources, especially in: Offshore Western India- our biggest hub for petroleum. Offshore Eastern India- Our primary source for natural gas. India is heavily dependent on coal (about 70%), which isn't great for the lungs of the planet. By mastering technology like CBM and Carbon Capture (CCS), India can stop seeing coal as just a "dirty" fuel and start seeing it as a tool for a cleaner, more secure energy future. The estimation of the future energy mix and the level of CO_2 emissions by 2030 is performed using a bottom-up energy-environmental optimization model, the AIM/Local model, which uses linear programming to determine the least cost combination of technologies to meet given demand and environmental targets and/or energy supply levels in a geographic area. Indian CO2 emissions in 2005 were 1229 Mt CO2, and almost doubled during 1990–2005. A significant fraction of the CO2 emanating from fossil-fuel combustion and specific industrial activities [11] was emitted by LPSs, with thermal power, steel, cement, fertilizer, petrochemical, aluminum plants, gas processing, and petroleum refineries contributing over 70% of all-India CO2 emissions in 2005. Coal contributed 72% of all-India CO2 emissions, overwhelmingly from power and steel plants.

METHODOLOGY:

CCO_2 Storage

The first step in using CCS technology to reduce CO_2 emissions is to identify opportunities to capture CO_2. The second part to look at is how to link captured CO_2 to other necessary criteria, such as potential storage for CO_2 around an LPS. CO_2 can be stored in deep geological formations for 1,000 to 1,000,000 years. Storage opportunities examined in this paper include sedimentary rock formations and deep unmineable coal seams.

CCS Clusters

There is potential for various CCS arrangements in India based on emitting source clusters. The majority of coal-fired plants are part of an industrial complex, which produces a large cumulative amount of CO_2. The capacity of the largest coal-fired plant in India exceeds 2000 MW. In addition, the planned ultra-mega coal-fired power plants will have a generating capacity of up to 4000 MW at the same site. The captured CO_2 will be transported to CO_2 storage sites via pipelines. Currently, Indian coal power plants mainly use sub-critical pulverized coal (PC) technology, which suits Indian coals well. The current standard for coal-power technologies in India is the BHEL 500 MW sub-critical PC unit. These units have assisted circulation boilers with a net steam pressure of 170 kg/cm^2. The best power plants, which are 500 MW sub-critical units, achieve a net efficiency of about 33%. In comparison, the average net efficiency for the 50 most efficient U.S. coal-based power plants is 36%, while the fleet average is 32%.

The average net efficiency for all coal power plants in the country is around 29%. Older units, those less than 200 MW, show the worst efficiencies. Despite low efficiencies and low plant load factors (PLF), these power plants keep running because they provide electricity at low costs. Therefore, improving the efficiency of existing power plants is vital for cutting greenhouse gas emissions. However, sub-critical PC technology does not meet many of India's future challenges, making it essential to explore other technological options. PC technology is becoming more efficient by using supercritical and ultra-supercritical steam properties, which require special materials. Supercritical technology is especially relevant for India in the short term. However, coal washing may be necessary to lower ash content when using supercritical boilers in India. The high ash content in Indian coals could lead to increased erosion of boiler tubes. In fact, the future of ultra-supercritical PC technologies might be limited by this high ash content. Currently, two supercritical NTPC plants based on Korean and Russian technologies are being built at Sipat and Barh, and they are expected to start operating by 2009. The CEA projects that about 17% (8 MW) of the new planned coal-based capacity for the 11th Plan will be supercritical PC. The government has proposed a new scheme called "Ultra-mega power plants." This plan includes at least seven supercritical plants of 4000 MW each, which will be under international bidding. These plants are expected to become operational after 2012, in the 12th Plan. In addition to improving efficiency, better post-combustion clean-up technologies like flue gas desulfurizers

(FGDs) and selective catalytic reducers (SCRs) can help reduce air pollution. Regarding CO_2 capture in PC plants, clean flue gas at atmospheric pressure with CO_2 concentrations below 15% can go through an amine-scrubbing process. This process selectively removes much of the carbon dioxide, which can then be compressed and transported for storage. However, capturing CO_2 in PC technologies will increase auxiliary consumption, coal use, and generation costs. Post-combustion capture in new PC plants may lead to a 24–40% increase in auxiliary energy consumption. For new pre-combustion plants in new IGCC plants, the increase ranges from 14–25%. Installing CO_2 capture equipment on existing plants also has a high cost in terms of power and efficiency. For example, an estimated 65 MW (30%) is lost in a 210 MW unit, with a 30% decrease in efficiency when the unit uses monoethanolamine (MEA)-based post-combustion capture technology. Carbon capture could double power generation costs in India. These high costs, along with the loss in net power and efficiency, make high-efficiency power plants essential for any potential retrofitting for carbon capture. Furthermore, there is still a lot of uncertainty about what actions a new power plant can take to be “capture-ready. IGCC, unlike PC plants, has several advantages. These include improved efficiency from the combined cycle, reduced resource consumption, better environmental performance, lower costs for pollution control technologies, and easier carbon capture. However, there are also drawbacks. These include complexity, high upfront costs, potentially lower reliability and availability, and less developed technology, which raises concerns about technology risk. In IGCC plants, carbon capture happens before combustion. The generated coal gas goes to a CO_2 shift reactor, where carbon monoxide turns into CO_2 and hydrogen (H_2) through a water gas shift reaction. CO_2 can then be separated using physical or chemical solvents. Before reaching the water shift reactor, other impurities like particulates and hydrogen sulfide must be removed from the coal gas. Overall, for bituminous coals, carbon capture with IGCC is more cost-effective than using amine-based capture in PC plants.

Coal is a sedimentary rock formed under elevated pressure and temperature for geologically long periods. This results in coal displaying a dual-porosity structure, consisting of both macropores (size > 50nm) and micropores (size < 2.0 nm) . A difference in pore size also affects the flow of materials (such as CO2), resulting in the application of different laws while studying such differential flows through the same medium. Gases follow Fick's law while diffusing through the micropores, and they follow Darcy's law while flowing through the coal cleats. CO2 injected into a coal body can stay in the cleats and micropores and lead to recovery of coal bed methane. Abandoned deep (> 500 m) or ultra-deep (> 800 m) mines that may be difficult to explore coal can be good targets for CCS.Studies by Vishal et al. (2017a) and Vishal et al. (2017b) concluded that coal permeability is lower for CO2 in a supercritical phase than in its liquid phase. The reason can be attributed to the high affinity of coal for the supercritical variety. To overcome this, injection pressure induced fracturing can be used to

agenda. The domestic and international steps laid out in this paper will help the Indian power sector develop a deliberate and careful approach pursued with short- and long-term perspectives in mind counteract the swelling. However, it should be done in a controlled way such that the fracture does not propagate throughout the entire coal seam.

CONCLUSION:

Although there are more immediate challenges for the power sector—especially the need to affordably provide more electricity to more people—GHG control and reduction of emissions from the coal-power sector will also need to be considered. The progress on a fair international climate regime will be critical for the implementation of deep carbon reduction strategies in India, and moreover, commitment and action by industrialized countries will be critical in catalyzing new research that reduces prices of carbon reducing technologies. In the meantime, India can implement a number of policy options that start reducing the GHG-emissions growth of the power sector while advancing the country's development.

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